IEEE Guide for Testing Faulted Circuit Indicators

IEEE Power Engineering Society

Sponsored by the
Insulated Conductors Committee

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IEEE-SA Standards Board

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Abstract: Definitions, service conditions, test procedures, and test conditions for faulted circuit indicators for use on power distribution systems are established in this test code.
Keywords: cable fault locator, fault indicators, faulted circuit indicator, FCI

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Introduction

This introduction is not part of IEEE Std 495-2007, IEEE Guide for Testing Faulted Circuit Indicators.

This guide is intended as a supplement to the training in the use of high-voltage electrical equipment, established safe operating procedures, and the manufacturer's instructions for the application of faulted circuit indicators (FCIs). Installers and operators of FCIs require formal training in the use of high-voltage electrical equipment. It is the responsibility of the users to establish safe operating procedures and provide training. The manufacturers are required to provide installation and operating instructions for their products.

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IEEE Guide for Testing Faulted Circuit Indicators

1. Overview

Faulted circuit indicators (FCIs) used by the electric utility industry are applied to overhead and underground power distribution circuits. This guide describes the general test requirements applied to FCIs.

1.1 Scope

This test code establishes definitions, service conditions, test procedures, and test conditions for FCIs for use on power distribution systems.

1.2 Purpose

FCIs are commonly used by electric utilities to aid in service restoration. The purpose of this guide is to identify the various tests and test conditions that FCIs must meet to validate their performance.

~~**References**~~

~~[1]ANSI / IEEE C37.41-1981, IEEE Standard De- sign Tests for High-Voltage Fuses, Distribution Enclosed Single-Pole Air Switches, Fuse Disconnecting Switches, and Accessories.'~~

~~[2] ANSI/IEEE Std 4-1978, IEEE Standard Techniques for High Voltage Testing.~~

2. Definitions

For the purposes of this guide, the following terms and definitions apply. *The Authoritative Dictionary of IEEE Standards Terms* [B5][1] should be referenced for terms not defined in this clause.

2.1 reset condition: The state of or the act of change of a faulted circuit indicator (FCI) indicating an unfaulted state.

2.2 reset device: A tool for manually resetting a faulted circuit indicator (FCI) from a faulted condition to an unfaulted condition.

~~**fault current.** Any current through the sensor equal to or in excess of the trip current of the faulted circuit indicator (FCI).~~

~~**faulted circuit indicator.** A single or multiphase device designed to sense fault current and provide an indication that the fault current has passed through the power conductor(s) at the point where the FCI sensor is installed.~~

~~ANSI documents are available from the Sales Depart ment, American National Standards Institute, **1430** Broad way, New York, NY 10018.~~

~~**indicator.** That portion of the FCI which indicates that fault current has been sensed.~~

~~**reset current or voltage.** The nominal rms value of current or voltage that will cause the indicator of the automatic current or voltage reset FCI to change from FAULT to NORMAL indication.~~

~~**reset time.** The time required for the FCI to return automatically to NORMAL indication after its reset current or voltage has been established, or for the elapsed time automatic reset FCI to reset.~~

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~~**response time.** The time required for the FCI to respond to a specified value of fault current.~~

~~**suitable test.** Where a condition or a set of conditions are ***so*** variable from one utility to another **or** even within the utility itself that no test can be properly specified for all , it is left to the user to determine their individual test needs. A suitable test and anticipated service life shall be that which is mutually agreed to between manufacturer and user.~~

~~**trip current.** The actual value of current in amperes rms that will cause the FCI to indicate FAULT.~~

~~**trip current rating.**~~

3. Service conditions

3.1 Usual service conditions

The FCI should be suitable for use under one or more of the following intended service conditions when:

- Exposed to direct sunlight salt spray, or normal expected pad mount conditions
- Not exposed to direct sunlight or salt spray
- Buried in earth (sensor only)
- Submerged in water intermittently or continuously to a depth of 4.5 m (15 ft)
- Exposed to rain, snow, hail
- Operating temperature range of –40 °C to +85 °C
- Applied on systems up to 46 kV phase-to-phase

3.2 Unusual service conditions

Service conditions other than those listed in 3.1 are considered to be unusual. Consult the manufacturer for recommendations.

4. Testing

4.1 Design tests

All applicable tests specified in 4.4 should be performed by the manufacturer to demonstrate that FCIs meet their ratings and are suitable for operation under the intended service conditions.

In addition, the manufacturer should periodically test a sufficient number of production FCIs to ensure continuing compliance with design tests.

4.2 Production tests

The production tests of 4.2.1 and 4.2.2 should be performed on every FCI.

4.2.1 Trip current rating verification test

~~see **4.4.9**~~

The trip current rating verification test should be performed by passing current through the conductor(s) to which the FCI is applied. The results should demonstrate compliance with the manufacturer's specified trip current rating and tolerances. This test should be performed at least one time at a temperature between 18 °C and 30 °C.

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4.2.2 Reset verification test

~~see **4.4.10**~~

The reset verification test should be performed to ensure that the FCI returns to a normal state, demonstrating compliance with the manufacturer's specified reset rating.

One of the following tests should be performed at least one time at a temperature between 18 °C and 30 °C:

a) *Current reset:* For FCIs that require a current flow through the cable conductor in order to change from an indication of FAULT to an indication of NORMAL, the following should apply: the indicator should change to an indication of NORMAL when the current specified by the manufacturer flows through the cable conductor for not more than 10 min.

b) *Voltage reset:* For FCIs that require a potential to be restored to the cable conductor in order to change the indicator from an indication of FAULT to an indication of NORMAL, the following shall apply: the indicator should move to an indication of NORMAL when the minimum reset voltage is applied to the cable conductor for not more than 10 min.

c) *Time reset:* For FCIs requiring only elapsed time to reset, the FCI should reset within its rated time and tolerances. The reset time tolerance is ± 20% of the manufacturers stated value. Since these times may be extremely long, it is not practical to perform this test in normal production. The manufacturer should demonstrate compliance during design testing.

d) *Manual reset:* For FCIs that require a manual operation to change from an indication of FAULT to an indication of NORMAL, the following shall apply: the indicator should move from the FAULT to NORMAL indication when reset by the means specified by the manufacturer.

~~The following production shall be performed by the manufactuer on every FCI:~~

4.3 General test conditions

The following test conditions should apply unless otherwise specified under specific tests:

a) Three samples should be tested unless otherwise agreed upon by the customer and the manufacturer of the device.

b) FCIs should be properly assembled with production grade components. All parts, which are grounded in a normal installation, shall be connected to the ground of the test circuit.

c) Test temperatures should be selected by the manufacturer to verify that the FCI meets its rating under the device's usual service conditions.

d) All ac voltages should be in accordance with IEEE Std 4™-1978 [B6].

e) The specific tests should be performed in the order as shown.

4.4 Specific design tests

4.4.1 Temperature cycling

The purpose of temperature cycling test is to ensure that the FCI operates after aging. The FCI should be subjected to five sequential thermal cycles at temperatures of –40 °C, +25 °C, and +85 °C. The following criteria shall be met unless otherwise agreed upon by the customer and the manufacturer: ~~Temperature equilibrium shall be achieved at each temperature level for the FCI before cycling can continue.~~

a) The temperature tolerance allowed in each case is ± 5 °C.

b) The humidity in the chamber should be between 0% and 90%, non-condensing.

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c) The temperature ramp rate should be no faster than 10 °C per hour.

d) The sequence should be +25 °C, +85 °C, +25 °C, −40 °C, +25 °C, and so on.

e) The dwell time at each temperature should be a minimum of 2 h.

Following these exposures, the FCI ~~shall~~ should pass all of the other applicable design tests.

4.4.2 Water submersion test

The purpose of the water submersion test is to verify that submersion cycling does not adversely affect the ability of the submersible FCI to prevent the entrance of moisture at all interfaces that could immediately or eventually render the device inoperable.

4.4.2.1 Submersion cycling

The following procedure shall be followed unless another procedure is agreed upon by the customer and manufacturer. The ramp rate for the thermal changes should be no faster than 10 °C per hour.

a) The FCI should be placed under water with an equivalent pressure head of 4.5 m (15 ft) ± 0.5 m (1.64 ft) of water or approximately 45 kPa (6.5 psi) at a temperature of +25 °C ± 5 °C for a minimum of 48 h.

b) The equivalent pressure head of 4.5 m (15 ft) ± 0.5 m of water or approximately 45 kPa (6.5 psi) should then be maintained as the water temperature is raised to +70° C ± 5 °C and held for a minimum of 48 h.

c) The equivalent pressure head of 4.5 m (15 ft) ± 0.5 m of water or approximately 45 kPa (6.5 psi) should then be maintained as the water temperature is allowed to return naturally to +25 °C ± 5 °C and held for a minimum of 2 h.

d) The equivalent pressure head of 4.5 m (15 ft) ± 0.5 m of water or approximately 45 kPa (6.5 psi) should then be maintained as the water temperature is lowered to a maximum of +5 °C and held for a minimum of 48 h.

e) The equivalent pressure head of 4.5 m (15 ft) ± 0.5 m of water or approximately 45 kPa (6.5 psi) should then be maintained as the water temperature is allowed to return naturally to +25 °C ± 5 °C and held for a minimum of 2 h.

4.4.2.2 Visual inspection

Following the submersion cycling tests of 4.4.2.1, the FCI should be removed from the water, cloth dried, and visually examined for moisture entrance at interfaces.

4.4.2.3 Post submersion cycling operation verification test

A normal production TRIP and RESET test as indicated in 4.2.1 and 4.2.2 may be performed to ensure that the submersion testing did not have an adverse affect on the device. It may be desirable to perform this test prior to proceeding with the remaining tests.

~~The FCI shall be placed under water with an equivalent pressure head of 15ft of water at a temperature of **20** °C to **30** °C for **48** h. The equivalent pressure head of **15** ft of water shall then be maintained while the water temperature is raised to **70** °C and held for **48** h. The equivalent pressure head of 15ft of water shall then be maintained as the water temperature is allowed to return naturally to 20 °C to **30** °C and while the water temperature is further lowered to **5** °C. This equivalent pressure head of 15ft of water and water temperature of 5°C shall be maintained for **48** h.~~

~~The FCI shall be removed from the water and visually examined for moisture inside the~~

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~~indicator and sensor and for any surface deterioration. The temperature of the FCI shall be allowed to return to 20°C to **30°C** naturally where it shall remain for **3** h. The tests of **4.4.9** and **4.4.10** shall than be applied. The temperature of the FCI shall be raised to 85"C where it shall remain for **3** h. The temperature of the FCI shall be held at 85"C and the tests of **4.4.9** and **4.4.10** shall again be applied. The temperature of the FCI shall then be lowered to -20 "C where it shall remain for **3** h. The temperature of the FCI shall be held at -20 "C and the tests of **4.4.9** and **4.4.10** shall again be applied.~~

~~* Numbers in brackets correspond to those of .the references, Section 1.2 of this standard.~~

~~**TESTING FAULTED CIRCUIT INDICATORS**~~

4.4.3 Outdoor weathering of plastics test

A suitable test should be performed to demonstrate that FCIs installed in direct sunlight perform satisfactorily for their anticipated service life.

4.4.4 Salt spray test

A suitable test should be performed to demonstrate that FCIs installed where exposed to salt spray perform satisfactorily for their anticipated service life.

4.4.5 Electric cord pull-out test

The purpose of the electric cord pull-out test is to ensure that electric cord strain relief used for auxiliary contacts, multiple part FCIs, or any other purpose can be accomplished so that a force exerted on the flexible cable is not transmitted to wiring connections. The assembly of a cord to an FCI should be capable of withstanding a steady-state straight pull equivalent to 13.6 kg (30 lb).~~The detachment of any cord conductor or any other evidence of mechanical or electrical failure is not acceptable.~~

a) The FCI should be secured in place with no less than a 13.6 kg (30 lb) mass suspended from the cord. This should remain in place for a minimum period of 1 h at +25 °C ± 5 °C. Note that other suitable automated test equipment may be used as long as the 13.6 kg (30 lb) minimum is applied.

b) Following the test, the FCI and cord should be examined to determine if there was any damage to the connections.

~~**4.4.5** Immersion Corrosion Test. A suitable test shall be performed to demonstrate that FCI's installed submerged in water or buried in earth will perform satisfactorily for their anticipated service life.~~

4.4.6 Impact resistance test

A suitable test should be performed to demonstrate that FCIs do not change state due to vibration or impact.

~~The purpose of this test is to ensure that the FCI will not change an indication from NORMAL to FAULT or from FAULT to NORMAL due to normal handling in the field. The FCI shall be mounted per the manufacturer's instructions on number **2 15** kV class concentric neutral cable connected with a 200 A class elbow to a **25** kVA low-profile padmounted transformer with an upward opening hinged lid. The FCI shall not change an indication from FAULT to NORMAL or from NORMAL to FAULT when the transformer lid is slammed open or shut.~~

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4.4.7 Short-time current test

The purpose of the short-time current test is to verify that the FCI is capable of withstanding short-time current of one of the magnitude and duration classes. See Table 1.

Table 1—Magnitude and duration classes

Fault current rating (A)	Test current (A) symmetrical	Time (s)	Minimum asymmetry factor
10 000	10 000	0.17	1.3 (X/R = 6)
25 000	25 000	0.17	1.6 (X/R = 20)

The rms value of the first major loop of the current wave should be equal to or greater than the symmetric value specified in Table 1, times the appropriate asymmetry factor. The rms value of the symmetric current should be equal to or greater than the value specified in Table 1. This short-time current test should be applied at ambient temperature in the range of +25 °C ± 10 C°. The test should be applied to the FCI two times.

4.4.8 Effect of current from adjacent conductors

The purpose of the following test is to verify that the FCI continues to indicate NORMAL (untripped) when the FCI is positioned in accordance with the manufacturer's recommended orientation and distance to an adjacent shielded conductor (underground) or unshielded conductor (overhead). The adjacent conductor should carry fault current as specified in one of the magnitude and duration classes. See Table 1. The temperature range should be +25 °C ± 10 °C.

a) The distance should be per the FCI specification or rating. It should be the minimum phase distance specified by the manufacturer in the FCI documentation.

b) The FCI is to be placed on a conductor energized to the requirements necessary for the device to function. This varies between types and manufacturers.

c) Energize the adjacent parallel conductor per one of the magnitude and duration classes as given in Table 1.

d) Verify that the FCI does not change from an indication of NORMAL (untripped) to an indication of FAULT (tripped).

e) This test is to be performed two times.

4.4.9 Trip current test

The purpose of the following trip current test is to verify that the FCI changes from an indication of NORMAL to an indication of FAULT:

a) Place the FCI on correct cable size for the design to be tested and, if necessary, energize the conductor per the product requirements for proper operation.

b) The test should be performed by passing current through the conductor(s) to which the FCI is applied. The results should demonstrate compliance with the manufacturer's specified trip current rating within tolerances.

c) The test should be run a minimum of two times at each of the temperatures: –30 °C ± 10 °C, +20 °C ± 10 °C, and +70 °C ± 10 °C. The FCI specimens are to dwell at each of the temperatures for a minimum of 1 h prior to performing the test.

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NOTE—The ability to run this test in a thermal chamber may not be possible in all cases. As an alternative, the FCI specimens can be conditioned in and then removed from the thermal chamber and tested, provided that the tests are completed within 15 min of removal from the chamber. Additionally, if this method is to be used, the FCI specimens should be allowed to dwell at each of the temperatures for a minimum of 2 h prior to removal and test.[2]

4.4.10 Reset test

The purpose of the following reset test is to verify that the FCI changes from an indication of FAULT to an indication of NORMAL

a) Place the FCI on correct cable size for the design to be tested.

b) Determine the type of FCI specimen and, following the manufacture's guideline for reset, apply any necessary voltage or current to verify compliance to the manufacture's ratings and specifications. Perform the proper test as indicated in 4.2.2 over the temperatures listed in step c).

c) The test should be run a minimum of two times at each of the temperatures: –30 °C ± 10 °C, +20 °C ± 10 °C, and +70 °C ± 10 °C. The FCI specimens are to dwell at each of the temperatures for a minimum of 1 h prior to performing the test.

NOTE—The ability to run this test in a thermal chamber may not be possible in all cases. As an alternative, the FCI specimens can be conditioned in and then removed from the thermal chamber and tested, provided that the tests are completed within 15 min of removal from the chamber. Additionally, if this method is to be used, the FCI specimens should be allowed to dwell at each of the temperatures for a minimum of 2 h prior to removal and test.

4.4.11 Time current test

The purpose of the time current test is to demonstrate that FCIs respond to trip magnitude overcurrent conditions within the manufacturer's specified trip response time. This test should be run and data recorded in a manner mutually agreed upon to demonstrate satisfactory response to user circuit protective devices.

[2] Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

~~**(1)** ***Trip Current Rating Verification Test.*** This test shall be performed by passing current through the conductor(s) to which the FCI is applied. The results shall demonstrate compliance with the manufacturer's specified trip current rating and tolerances.~~

~~The manufacturer shall perform this trip current test at various temperatures throughout the specified temperature range of the FCI with the selection designed to demonstrate compliance with the requirements of this test code. This test shall be performed a minimum of ten times at each temperature selected for design tests and once at 20°C to 30°C for production tests.~~

~~***(2) Test for Effect of Current Adjacent Conductors.*** This test shall verify that the indicator will continue to indicate NORMAL when he FCI is positioned in any orientation no closer than the manufacturer's specified distance to an adjacent unshielded conductor carrying fault current **as** specified in either the **10000A** or **25** 000 A class in **4.4.8.** The temperature shall be the same as **4.4.9 (1).**~~

~~**4.4.10** Reset Test. If this test is being conducted **as** a design test, the specimens shall first have passed the short-time current test, **4.4.8. (1)** ***Automatic Reset FCI's.***~~

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Annex A

(informative)

Other considerations

Other factors that may need to be considered when testing FCIs are as follows:

a) This guide does not address battery or capacitor testing. The working group plans to include this information in future revisions of this guide.

b) Due to the wide variety of types and operational techniques, it may be necessary to consult the manufacture of the device for assistance with test methods. Variations of the test methods described in this guide are only to be made if agreed upon by the customer and the manufacturer.

c) The FCI only needs to be tested for the area of intended use.

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Annex B

(informative)

Glossary

fault current: A current that flows on a circuit due to short-circuit conditions.

faulted circuit indicator (FCI): A single- or multi-phase device designed to sense fault current and provide an indication that the fault current has passed through the power conductor(s) at the point where the FCI sensor is installed.

indicator: That portion of the FCI that indicates that fault current has been sensed.

inrush restraint: An FCI design feature to minimize false tripping due to current inrush during energization of the circuit.

logic circuit: An FCI internal circuit design for recognizing a fault condition. The section of the FCI that determines if and when a fault condition exists.

manual reset faulted circuit indicator (FCI): A type of FCI that requires manual reset by the operator.

proximity effect: The magnetic induction effect of load or fault current flowing in an adjacent wire, cable, or ground conductor that may cause an FCI to malfunction (i.e., false trip, failed to trip, or reset incorrectly).

reset current or voltage (FCI): The nominal rms value of current or voltage that will cause the indicator of the automatic current or voltage reset FCI to change from FAULT to NORMAL indication.

reset time: The time required for the FCI to return automatically to NORMAL indication after its reset current or voltage has been established, or for the elapsed time automatic reset FCI to reset.

remote display: An FCI that is physically separated from the sensor.

response time: The time required for the FCI to respond to a specified value of fault current.

sensor: The sensing section of an FCI designed to detect a fault condition.

suitable test: Where a condition or a set of conditions are so variable from one utility to another or even within the utility itself that no test can be properly specified for all conditions, it is left to the user to determine their individual test needs. A suitable test and anticipated service life are mutually agreed to between manufacturer and user.

time reset: A type of automatic reset that resets an FCI after a specified time.

trip current: The actual value of current in amperes rms that will cause the FCI to indicate FAULT.

trip current rating: The published rms sinusoidal fault current in amperes that cause the FCI to indicate FAULT.

trip level: The threshold current or rate of rise that will cause the FCI to operate.

tripped faulted circuit indicator (FCI): An FCI that has operated to indicate a fault condition.

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